Mobile and Wireless Communication

Prepared by: Er. Grishma Poudel

BE (Electronics and Communication Engineering)

IOE, Thapathali Campus

ME (Master of Computer Engineering)

Pokhara University, NCIT (PU scholarship)

Course Evaluation

Evaluation:

Final -- 50

Sessional – 50

- Internal Exam – 40%
- Assignment -30%
- Attendance and Discipline - 10%
- Case Study and Group Discussion – 20%

Syllabus Overview

Mobile and Wireless Communication CMM420

Introduction

1.1 Definition, advantage and disadvantage of wireless communication System

1.2 Evolution of Mobile Radio Communications (1G , 2G, 3G, 4G, 5G….)

1.3 CDMA,GSM,DECT

Definition

It refers to the transmission of information over a distance without the need for wires, cables, or any other physical connection between sender and receiver.

Wireless communication is simply data communication without the use of a physical connection or wires but propagates the signal through space.

It technology typically use electromagnetic waves.

Some of the key applications of wireless communication include:

- Mobile Communication
- Wireless Internet
- Bluetooth
- Satellite Communication
- Wireless sensors
- Etc

    - *That was 1897 Since G.Marconi first demonstrated radio's ability to provide continuous contact with ships, from there the evolution of wireless technology. (T.S.Rappaport)*

Advantage and disadvantage of wireless communication System

Advantages of wireless communication are as:

- **Mobility**: Wireless communication allows users to communicate and access information from anywhere within the coverage area without being tighten to a specific location by cables or wires.

- **Convenience**: Wireless technology eliminates the need for physical connections, making it convenient to connect devices and access information without the hassle of cables.

- **Flexibility**: Wireless networks can be easily deployed and scaled to accommodate changing needs, making them suitable for dynamic environments and temporary setups.

- **Cost-Effectiveness**: . The elimination of physical cables and the associated labor costs for installation and repairs.

- **Network planning**: Network planning in wireless communication is very easy as compare to wired network due to wireless software configuration of frequency, power and other parameter.

- **Ease of Installation**: Setting up wireless systems can be faster and less invasive than traditional wired networks, which often require extensive cabling and infrastructure.

- **Scalability**: Wireless networks can easily accommodate a growing number of users and devices by adding additional access points or expanding coverage areas as needed.

Disadvantages

- As communication is done through open space, it is less secure.
- Unreliability
- More open to interference.
- Increased chance of jamming.
- Transmission speed is comparably less.
- it has a limited amount of bandwidth for communication and breaches of network security.

Evolution of Mobile Radio Communication.

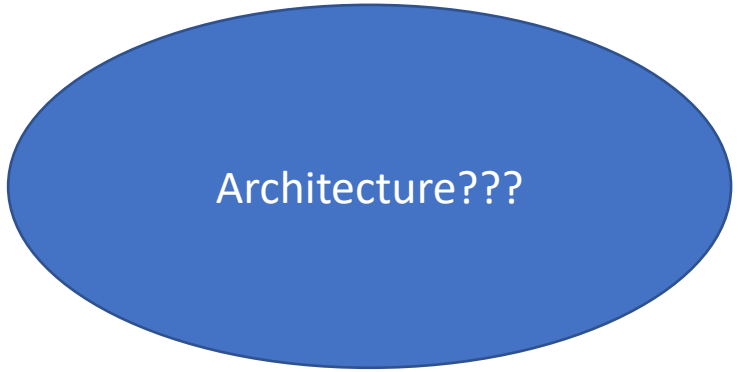
- 1946- First domestic public land mobile service introduction in st.Louis. The system operated at 150 MHz and has only three channels.
- 1956- First use of 450 MHz system. Users had to use a Push to talk (PTT) button and always needed operator assistance.
- 1970 FCC sets aside 75 MHz for high- capacity mobile telecommunication systems.
- 1974 FCC grants common carriers 40 Mhz for development of cellular systems.
- 1978 First cellular system called AMPS was introduced in Chicago on a trail basis.
- 1981 Cellular System deployed in Europe.

- 1983 First commercial deployed of cellular system in Chicago. It is an analog system and does not have a user data transport capability. Analog system around 450 and 900MHz band were also introduced in many countries of Europe during 1981-90.
- 1991 GSM introduced in Europe and other countries of the world.
- TDMA system called IS-54 introduce in the US. SMS available in GSM.
- 1995 CDMA cellular and PCS technology introduced in the US.
- 1997 ETSI publishes GPRS standard.
- 1999 Standards for 3G wireless services published.
- 2007 standards for 4G wireless serviced Published.
- 2019 5G was first introduced.

First-Generation Network

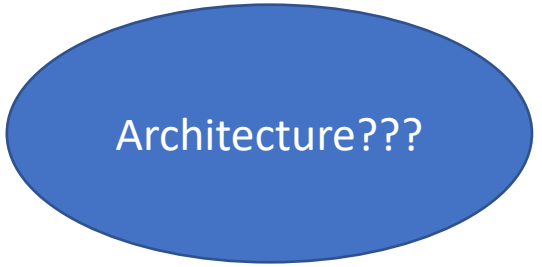
Developed: late 1970s and early 1980s

- Used analog signal for voice services
- Based on technology called as Advanced Mobile Phone System (AMPS)
- Frequency modulated and Frequency division multiple access (FDMA) with a channel capacity of 30KHz
- circuit switching
- Frequency band of 450MHz – 1GHz
- Allows voice calls in 1 country



Second Generation

- First introduced in 1990s.
- Data service Capability.
- Use digital modulation formats.
- Use TDMA/FDD and CDMA/FDD multiple access technology.
- Sophisticated digital signal processing in handsets and base stations.
- Three times increment in spectrum efficiency as compared to 1G.
- Connected to public data network (PDN).



for example, IS-95 supports circuit switched data and digital fax, IP mobile.

2.5 G Network

- 2G cellular technology with General Packet Radio Service (GPRS),
- Data rate upto 114 kbps
- Enhanced Data Rates for GSM Evolution (EDGE), providing 384 kbps rate by using improved modulation (8-PSK instead of GMSK in GSM)
- CDMA: high speed data access multiple orthogonal user channels providing 115.2kbps rate
- Support email, web browsing
- Enabling location - based mobile service

3G Network

Foundation of mobile broadband

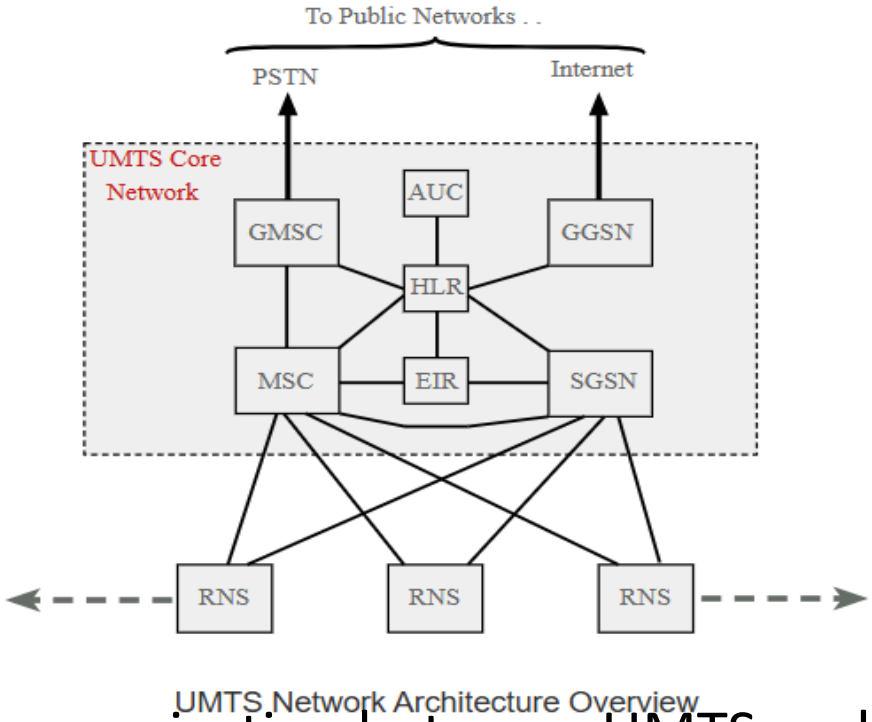
- Developed: 2000 based on GSM
- Support Voice, SMS, Data
- Technology: UMTS, WCDMA and FDD/TDD
- Data rate 144 kbps to 2 Mbps enable MMS
- Digital modulated (QPSK, QAM)
- Circuit switching for voice and packet for data
- Frequency band of 1.6GHz – 2GHz

UTMS (Universal Mobile Telecommunication System)

GMSC(Global Mobile Switching Center):- Provides Interfaces to PSTN or other mobile networks.

AUC (Authentication Center): It responsible to the Security and authentication of mobile subscriber accessing the network.

GGSN(Gateway GPRS support Node): The GGSN is a key component of the UMTS core network architecture, specifically within the packet-switched domain, and it plays a crucial role in enabling data communication between UMTS mobile devices and external packet-switched networks, such as the internet or corporate intranets.



UMTS Network Architecture Overview

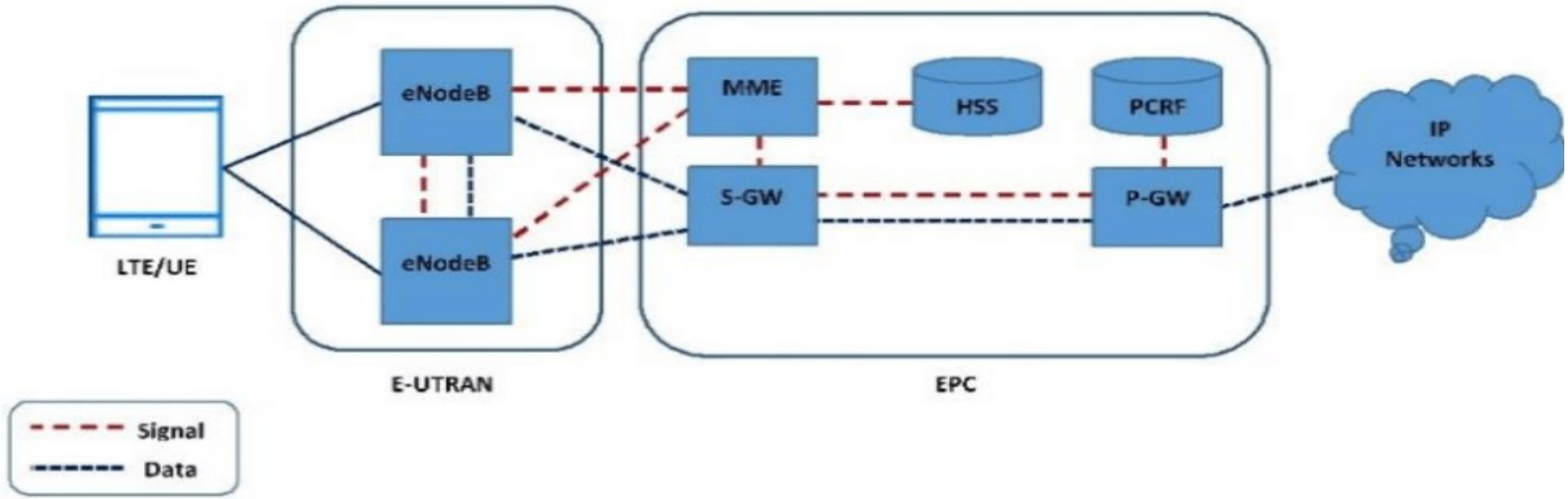
SGSN(Serving GPRS support Node): Mobility management, Packet Routing, IP Allocation

RNS (Radio Network Controller): Radio Resource management, Connection establishment and release etc.

4G network

Developed: 2010

- Support Voice, SMS, Data (High Speed)
- Technology: long term evolution (LTE), OFDMA and FDD/TDD
- Data rate 100 Mbps to 1 Gbps enable MMS
- Digital modulated (QAM)
- Packet switching
- Frequency band of 2GHz – 8GHz
- Good QoS and high security
- Bigger battery usage



4G Network Architecture

MME (Mobility Management Entity): Responsible for session management and mobility. It manages signaling between the UE (User Equipment) and the core network.

SGW (Serving Gateway): Serves as the local mobility anchor, routing the data packets.

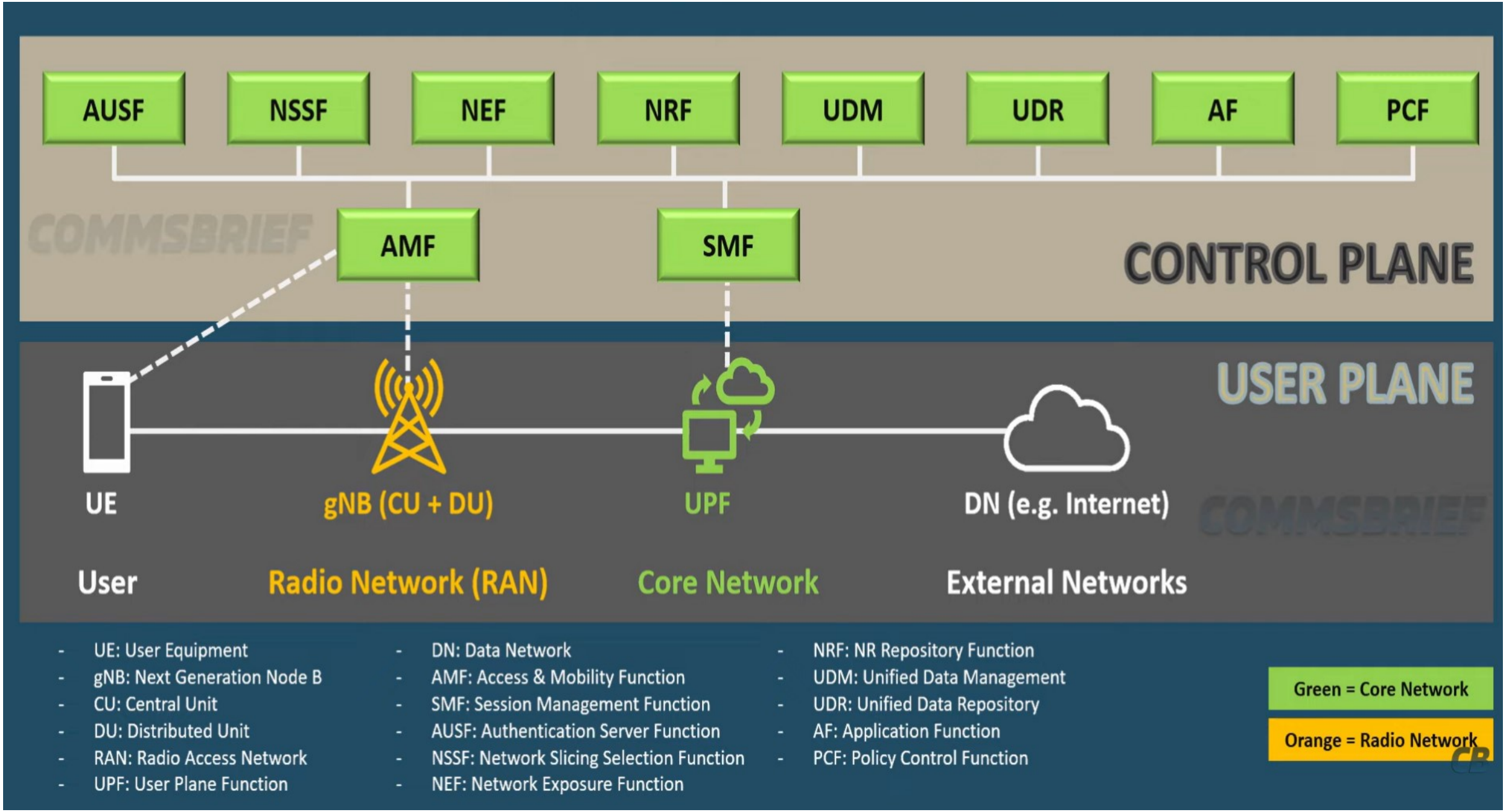
PGW (PDN Gateway): Manages IP data connections and assigns IP addresses to UEs.

HSS (Home Subscriber Server): A central database for user information and subscriptions.

PCRF (Policy and Charging Rules Function): Dictates policy and charging rules.

5G Network

- Developed: 2020
- main focus :world-Wireless World Wide Web (WWWW)
- Technology: NOMA, mmWave, small cell, MIMO
- Data rate > 1Gbps High speed, high capacity
- Internet switching


AUSF
NSSF
NEF
NRF
UDM
UDR
AF
PCF
AMF
SMF
CONTROL PLANE
USER PLANE
UE
gNB (CU + DU)
UPF
DN (e.g. Internet)
User
Radio Network (RAN)
Core Network
External Networks
- UE: User Equipment
- gNB: Next Generation Node B
- CU: Central Unit
- DU: Distributed Unit
- RAN: Radio Access Network
- UPF: User Plane Function
- DN: Data Network
- AMF: Access & Mobility Function
- SMF: Session Management Function
- AUSF: Authentication Server Function
- NSSF: Network Slicing Selection Function
- NEF: Network Exposure Function
- NRF: NR Repository Function
- UDM: Unified Data Management
- UDR: Unified Data Repository
- AF: Application Function
- PCF: Policy Control Function
Green = Core Network
Orange = Radio Network
COMMSBRIEF

Network Slice Selection Function (NSSF):NSSF plays a crucial role in enabling this flexibility by facilitating the dynamic selection and allocation of network slices based on the needs of users and services.

Network Exposure Function (NEF): NEF provides APIs for external applications to interact with the 5G network.

Network Repository Function (NRF):NRF is a crucial element in 5G networks responsible for maintaining a centralized repository of network functions available within the network.

Policy Control Function (PCF):PCF is a critical component in the 5G core network responsible for enforcing policies related to quality of service (QoS), network resource allocation, and traffic management.

Unified Data Management (UDM): UDM is a core network function in 5G architecture responsible for storing and managing user subscription data and profiles.

Application Function (AF): The Application Function is a key component in the 5G architecture responsible for supporting service-specific applications and interacting with other network functions to enable services.

Authentication Server Function (AUSF): AUSF is a crucial component in the 5G core network responsible for authenticating and authorizing users and devices accessing the network.

Access and Mobility Management Function (AMF): AMF is a key component in the 5G core network responsible for managing access to the network and handling mobility-related functions for user devices.

Session Management Function (SMF): SMF is a crucial component in the 5G core network responsible for managing user sessions and the associated data flows.

CDMA

Code Division Multiple Access (CDMA) is a digital cellular technology that allows multiple users to share the same frequency band simultaneously by assigning unique codes to each user. Here's how it works:

- **Spread Spectrum Technique:** CDMA uses a spread spectrum technique, where each user's data is spread across the entire available bandwidth using a unique code. These codes are orthogonal to each other, meaning they have minimal interference with each other when transmitted simultaneously.
- **Unique Code Assignment**: Each user in a CDMA system is assigned a unique code, known as a spreading code or pseudo-random noise (PN) code. These codes are typically long sequences of 1s and 0s, which are carefully designed to have desirable properties such as low cross-correlation and high auto-correlation.
- **Decoding Process:** At the receiver side, each user despreads the received signal using their assigned spreading code. The receiver correlates the received signal with its own spreading code, effectively extracting the desired user's data from the received signal.

Advantages of CDMA:

- **Increased capacity**: CDMA allows multiple users to share the same frequency band simultaneously, effectively increasing the capacity of the system.
- **Improved security:** Since each user's data is encoded with a unique spreading code, it is difficult for unauthorized users to intercept or decode the transmitted data.
- **Better resistance to multipath fading:** CDMA signals exhibit robustness against multipath fading, where signals take multiple paths due to reflection and scattering, making it suitable for urban and indoor environments.
- **Flexible allocation of resources:** CDMA systems can dynamically allocate resources based on user demand, optimizing the utilization of available bandwidth.

GSM

GSM, which stands for Global System for Mobile Communications, is a widely used digital cellular technology that revolutionized mobile communication.

Frequency Division Multiple Access (FDMA): GSM divides the available frequency spectrum into multiple channels, each of which is allocated to a different user. FDMA is used to separate users by frequency. Each user is assigned one frequency channel for the duration of the call.

Time Division Multiple Access (TDMA): Within each frequency channel, GSM further divides the transmission into time slots. TDMA is used to divide each frequency channel into multiple time slots, allowing multiple users to share the same frequency by transmitting their signals in different time slots.

GSM

GSM networks consist of several key components, including:

Mobile Station (MS): This includes the mobile device (phone) and the Subscriber Identity Module (SIM) card.

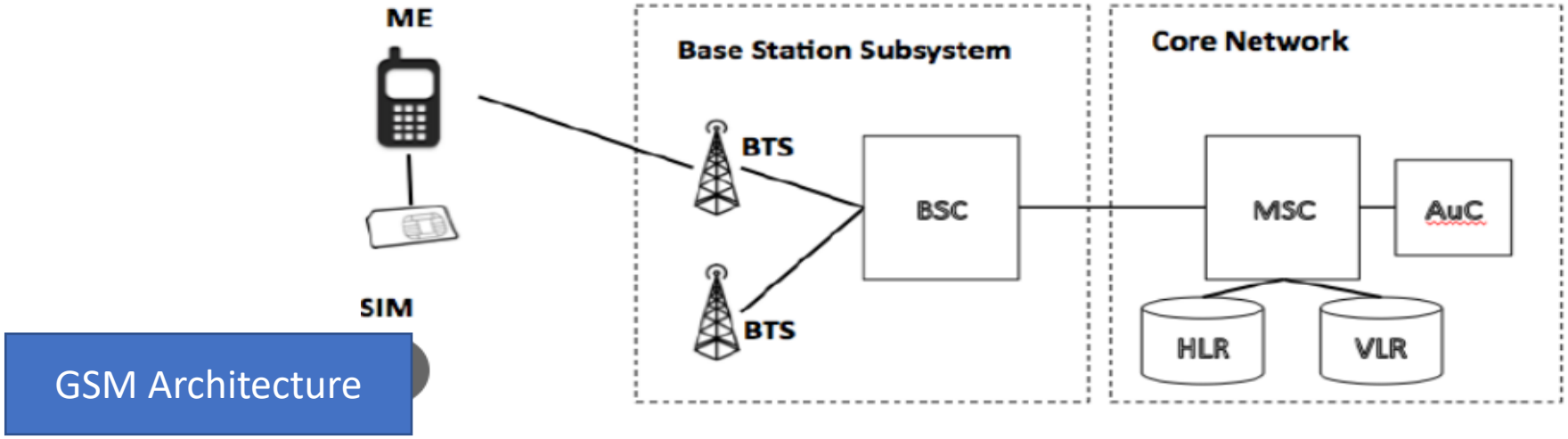
Base Station Subsystem (BSS): This comprises the Base Transceiver Station (BTS) and the

Base Station Controller (BSC). The BTS communicates directly with mobile devices, while the BSC manages multiple BTSs.

Network Subsystem (NSS): This consists of the Mobile Switching Center (MSC) and the

Home Location Register (HLR). The MSC handles call routing and switching, while the HLR stores subscriber information.

Operation and Support Subsystem (OSS): This includes the components necessary for network management and maintenance.



GSM Architecture

DECT

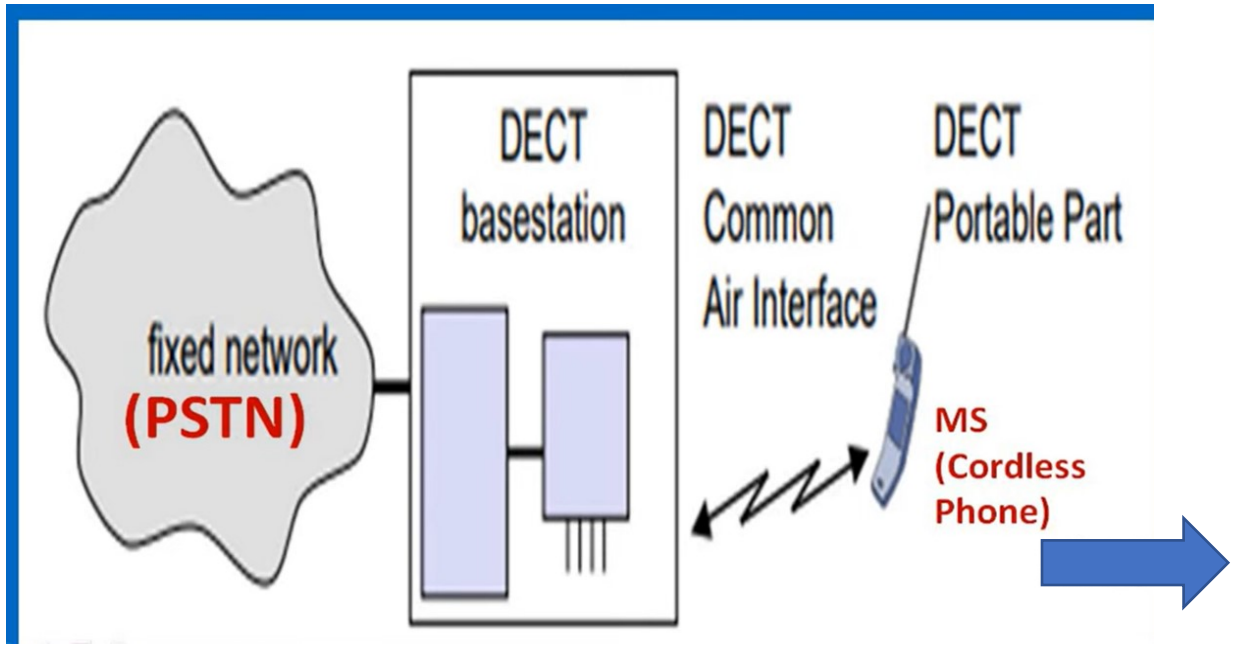
DECT, which stands for Digital Enhanced Cordless Telecommunications, is a wireless communication technology primarily used for cordless phones, wireless home and office telephony, and other short-range communication applications. Here's an explanation of how DECT works:

- **Frequency Bands**: DECT operates in the frequency bands between 1880 and 1900 MHz in most countries. DECT also uses Frequency Division Duplexing (FDD), where different frequencies are used for the uplink (from the handset to the base station) and downlink (from the base station to the handset) transmissions.
- **Time Division Multiple Access (TDMA)**: DECT utilizes Time Division Multiple Access (TDMA) to divide the available frequency band into time slots.
- **GAP Compatibility:** DECT devices are often designed to be compatible with the Generic Access Profile (GAP), a standard that enables interoperability between different manufacturers' DECT products.
- **Encryption and Security:** DECT employs encryption to secure voice communications between the handset and the base station.

DECT

- **Range and Coverage:** DECT offers a range of up to 300 meters (approximately 984 feet) indoors and up to 1000 meters (approximately 3280 feet) outdoors, depending on environmental factors and obstacles.

Overall, DECT technology offers reliable, secure, and high-quality wireless communication for various short-range applications, making it a popular choice for cordless telephony and other wireless communication needs.





Comparison between CDMA, GSM and DECT

- **Multiple Access Techniques:**

CDMA: Utilizes Code Division Multiple Access, spreading signals across the entire available bandwidth using unique codes.

GSM: Utilizes both Frequency Division Multiple Access (FDMA) and Time Division Multiple Access (TDMA) to separate users by frequency and time slots.

DECT: Utilizes TDMA to divide the frequency band into time slots, allowing multiple devices to share the same frequency by transmitting their signals in different time slots.

- **Frequency Bands:**

CDMA: Operates in various frequency bands, depending on the specific implementation (e.g., 800 MHz, 1900 MHz).

GSM: Operates in the frequency bands around 900 MHz and 1800 MHz.

DECT: Operates in the frequency bands between 1880 and 1900 MHz, reserved exclusively for DECT communication.

- **Architecture:**

CDMA: Typically used in cellular networks with components such as Base Transceiver Stations (BTS) and Mobile Switching Centers (MSC).

GSM: Also used in cellular networks with components like Base Transceiver Stations (BTS), Mobile Switching Centers (MSC), and Home Location Registers (HLR).

DECT: Primarily used for cordless telephony and short-range communication, consisting of handsets and base stations.

- **Encryption and Security:**

CDMA: Provides encryption for voice and data transmissions, enhancing security.

GSM: Implements encryption (A5 encryption algorithm) for voice and data transmissions, ensuring confidentiality.

DECT: Employs encryption (DECT Standard Cipher) to secure voice communications between handsets and base stations, providing confidentiality and protection against eavesdropping.

- **Range and Coverage:**

CDMA: Typically offers coverage similar to other cellular technologies, depending on the deployment and infrastructure.

GSM: Offers coverage similar to CDMA, with range and coverage depending on the deployment and infrastructure.

DECT: Offers a range of up to 300 meters indoors and up to 1000 meters outdoors, suitable for cordless telephony and short-range communication.

Assignment for chapter 1

1. Explain the wireless system. How wireless is better than wired network.
2. Explain the advantage and disadvantage of wireless over the wired network.
3. Compare the different generation on network.
4. Explain the GSM system.
5. Explain the CDMA system.
6. Explain the DECT system.
7. Compare the GSM,CDMA and DECT technology.